NIRANGA NILUMINDA

Software Engineer / Backend & Distributed Systems

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Summary

Computer Engineering graduate (GPA 3.80/4.00, final results pending) who builds backend and distributed systems end to end: event-driven microservices on Kafka, Spark data pipelines, and security research combining graph neural networks, post-quantum cryptography and blockchain. All projects below are public on GitHub with tests and CI.

Education

University of Ruhuna – BSc (Hons) Computer Engineering 2022 – 2026 | **GPA 3.80 / 4.00**
All final examinations completed September 2026; results expected November 2026.

Technical Skills

Languages	Python, TypeScript/JavaScript, SQL, C++, Go, Solidity
Backend	Microservices (SAGA, Outbox, CDC), FastAPI, Node.js/Express, Next.js, React, REST, JWT/RBAC
Data	Apache Kafka (Avro, Debezium CDC), Spark (Structured Streaming, batch), Airflow, PostgreSQL, MongoDB
DevOps	Docker, GitHub Actions, Jenkins, SonarQube, Terraform, Ansible, AWS (EC2, IAM), Grafana, pytest, Vitest

Projects

Exam & Certification Platform / *Microservices, Kafka, Debezium, Docker* GitHub: exam-certification-platform

- Built 8 microservices (database per service) where enrolment, payment, grading and PDF certification are coordinated by an orchestrated **SAGA** over a 3-broker Kafka cluster, with compensating actions on failure.
- Removed dual-write risk with the **Transactional Outbox** pattern and Debezium CDC; secured all traffic through an RS256 JWT API gateway with role-based access control.
- CI/CD with GitHub Actions (build, SonarQube, GHCR) and Jenkins (staging, smoke tests, approval, production).

Smart Grid Lambda Platform / *Kafka, Spark, Airflow, FastAPI, Grafana* GitHub: smart-grid-lambda-platform

- Built a Lambda-architecture data platform: smart-meter streams through Kafka into Spark Structured Streaming (watermarks, sliding windows) for live metrics, plus an Airflow-orchestrated Spark batch layer for billing.
- Idempotent, recomputable batch layer with pluggable tariff strategies; FastAPI serving, Grafana alerting, **127 tests**.

SENTINEL: Attack Detection for Vehicular Networks / *C++, PyTorch, Fabric* GitHub: MPTD-PQS-SDVN

- Final-year research (team of 4, led architecture and most implementation): NS-3 + SUMO simulation detecting 7 attack variants from malicious vehicles, roadside units, man-in-the-middle and compromised controllers.
- Combined rule signatures with a Graph Attention Network and LSTM autoencoder (PyTorch to ONNX, run in C++), post-quantum threshold signatures (ML-DSA-87, OpenFHE) and Go smart contracts on Hyperledger Fabric; beat three published baselines on coordinated Sybil and control-plane attacks, validated by 11 ablation studies.

SecureOTA / *Solidity, Python, React, TypeScript* GitHub: secure-ota

- Technical lead (team of 4): IoT firmware updates verified against SHA-256 fingerprints on an Ethereum smart contract, blocking tampered, rolled-back and revoked firmware; architecture rules enforced in CI; 142 tests.

PrepDeck / *Next.js 16, TypeScript, MongoDB, Zod, Vercel* prep-deck-seven.vercel.app

- Live interview-prep app with spaced repetition, timed mock exams and community moderation; repository-pattern data layer keeps MongoDB swappable for PostgreSQL; **225 Vitest tests**.

TradeBot / *Python, SQLite, LLM APIs, systemd* GitHub: TradeBot

- Crypto trading bot with backtesting and walk-forward optimisation, an LLM news-risk gate, exchange testnet trading with stop orders, a Telegram and web dashboard, and restart-safe deployment as systemd services; **84 tests**.

Experience

MO Marketplace – Software Engineer | Client-side Product Owner (Intern) Colombo | May 2025 – Nov 2025

- Analysed the production PostgreSQL schema and wrote the SQL behind Apache Superset dashboards for user activity, product performance and seller metrics.
- Ran UAT and regression testing on every fortnightly mobile release; wrote PRDs and managed the Jira backlog between stakeholders and the engineering team.

Achievements

Top 10 national finalist, Agentrix 2026 | **Top 5 finalist**, Eminence (University of Ruhuna) | CodeSprint, CTF, CodeFest AI Hackathon, Haxtream | Volunteer District Coordinator, Mehewara Seminar Series

References

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